

#Atmospheric Dynamics



Chapter 3: The Equation Of Motion



Learning outcome

- *To Understand The Meaning Of Conservation*
- *To Be familiar with the laws of Conservation of mass*
- *To Understand the laws of Conservation of momentum*
- *To Perceive the laws of Conservation of energy*

What is a conservation law?

- ❑ In ordinary language, “conservation of X” usually means “please avoid wasting X”.
- ❑ In physics, however, the word “conservation” has a quite different meaning: “conservation of X” or “X is conserved” means that X is constant in time, i.e.

$$dX/dt = 0$$

Conservation Law

- ❑ Atmospheric motions are governed by three fundamental physical principles: *conservation of mass*, *conservation of momentum*, and *conservation of energy*.
- ❑ The mathematical relations that express these laws may be derived by considering the budgets of mass, momentum, and energy for an infinitesimal *control volume* in the fluid.
- ❑ Two types of control volume are commonly used in fluid dynamics.

The Laws Of Conservation Of Mass

THE CONTINUITY EQUATION

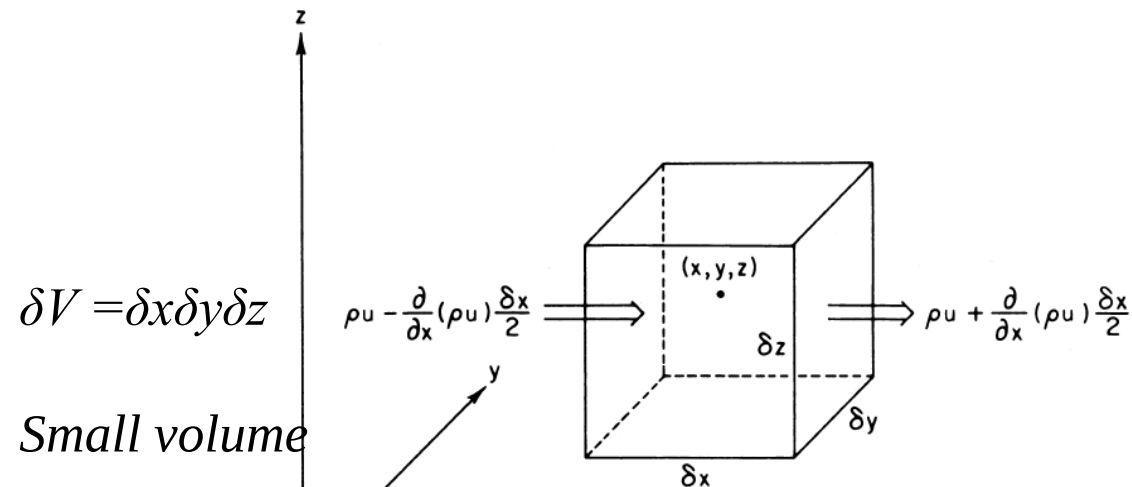
- ❑ The law of conservation of mass states that for any system closed to all transfers of matter and energy, *the mass of the system must remain constant over time*, as the system's mass cannot change, so quantity can neither be added nor be removed.
- ❑ Therefore, the quantity of mass is conserved over time.
- ❑ The law implies *that mass can neither be created nor destroyed*
- ❑ The mathematical relationship that expresses conservation of mass for a fluid is called the *continuity equation*.
- ❑ The continuity equation hold true An open system

The Continuity Equation

- *Continuity equation* is the mathematical relationship that expresses conservation of mass for a fluid.
- For a fixed control volume, the net rate of mass inflow through the sides must equal the rate of accumulation of mass within the volume

The rate of inflow of mass per unit area is:

$$\left[\rho u - \frac{\partial}{\partial x}(\rho u) \frac{\delta x}{2} \right]$$



The rate of outflow of mass per unit area is:

$$\left[\rho u + \frac{\partial}{\partial x}(\rho u) \frac{\delta x}{2} \right]$$

The Continuity Equation

- Area = $\delta y \delta z$
- The net rate of flow into the volume due to the x velocity component is:

$$\left[\rho u - \frac{\partial}{\partial x}(\rho u) \frac{\delta x}{2} \right] \delta y \delta z - \left[\rho u + \frac{\partial}{\partial x}(\rho u) \frac{\delta x}{2} \right] \delta y \delta z$$
$$= -\frac{\partial}{\partial x}(\rho u) \delta x \delta y \delta z$$

- Similarly, the net rate of flow of mass in y and z components:

$$-\frac{\partial}{\partial y}(\rho v) \delta x \delta y \delta z \quad \text{and} \quad -\frac{\partial}{\partial z}(\rho w) \delta x \delta y \delta z$$

The Continuity Equation

- Thus, the net rate of mass inflow is

$$-\left[\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) \right] \delta x \delta y \delta z$$

- the mass inflow per unit volume is just $-\nabla \cdot (\rho \mathbf{U})$, which is equal to the rate of mass increase per unit volume.
- The above equation is the mass divergence form of the continuity equation

The Laws Of Conservation Of Momentum

- ❑ The conservation law for momentum (Newton's second law of motion) relates the rate of change of the absolute momentum following the motion in an inertial reference frame to the sum of the forces acting the fluid.
- ❑ For most applications in meteorology it is desirable to refer the motion to a reference frame rotating with the earth.
- ❑ Transformation of the momentum equation to a rotating coordinate system requires a relationship between the total derivative of a vector in an inertial reference frame and the corresponding total derivative in a rotating system.

- ❑ The law of **conservation of momentum** in atmospheric dynamics implies that The momentum of a fluid element remains constant unless acted upon by external forces.

Total Differentiation in a Rotating System

- The conservation law for momentum relates the rate of change of the absolute momentum following the motion in an inertial reference frame to the sum of the forces acting on the fluid
- Transformation of the momentum equation to a rotating coordinate system requires a relationship between the total derivative of a vector in an inertial reference frame and the corresponding total derivative in a rotating system.

To derive this relationship:

Let $\vec{A}$ be an arbitrary vector with Cartesian components

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \quad \text{in an inertial frame of reference, and}$$

$$\vec{A} = A'_x \hat{i}' + A'_y \hat{j}' + A'_z \hat{k}' \quad \text{in a rotating frame of reference.}$$

Total Differentiation in a Rotating System

If $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ in an inertial frame of reference, then

$$\frac{d\vec{A}}{dt} = \left(A_x \frac{d\hat{i}}{dt} + \hat{i} \frac{dA_x}{dt} \right) + \left(A_y \frac{d\hat{j}}{dt} + \hat{j} \frac{dA_y}{dt} \right) + \left(A_z \frac{d\hat{k}}{dt} + \hat{k} \frac{dA_z}{dt} \right)$$

Since the coordinate axes are in an inertial frame of reference,

$$\frac{d\hat{i}}{dt} = \frac{d\hat{j}}{dt} = \frac{d\hat{k}}{dt} = 0$$

$$\frac{d\vec{A}}{dt} = \left(\cancel{A_x \frac{d\hat{i}}{dt}} + \hat{i} \frac{dA_x}{dt} \right) + \left(\cancel{A_y \frac{d\hat{j}}{dt}} + \hat{j} \frac{dA_y}{dt} \right) + \left(\cancel{A_z \frac{d\hat{k}}{dt}} + \hat{k} \frac{dA_z}{dt} \right)$$

$$\frac{d\vec{A}}{dt} = \frac{dA_x}{dt} \hat{i} + \frac{dA_y}{dt} \hat{j} + \frac{dA_z}{dt} \hat{k} \quad (\text{Eq. 1})$$

Total Differentiation in a Rotating System

If $\vec{A} = A'_x \hat{i}' + A'_y \hat{j}' + A'_z \hat{k}'$ in a rotating frame of reference, then

$$\frac{d\vec{A}}{dt} = \left(A'_x \frac{d\hat{i}'}{dt} + \hat{i}' \frac{dA'_x}{dt} \right) + \left(A'_y \frac{d\hat{j}'}{dt} + \hat{j}' \frac{dA'_y}{dt} \right) + \left(A'_z \frac{d\hat{k}'}{dt} + \hat{k}' \frac{dA'_z}{dt} \right) \quad (\text{Eq. 2})$$

Because the left hand sides of Eq. 1 and Eq. 2 are identical,

$$\frac{dA'_x}{dt} \hat{i}' + \frac{dA'_y}{dt} \hat{j}' + \frac{dA'_z}{dt} \hat{k}' = \left(A'_x \frac{d\hat{i}'}{dt} + \hat{i}' \frac{dA'_x}{dt} \right) + \left(A'_y \frac{d\hat{j}'}{dt} + \hat{j}' \frac{dA'_y}{dt} \right) + \left(A'_z \frac{d\hat{k}'}{dt} + \hat{k}' \frac{dA'_z}{dt} \right)$$

Regrouping the terms

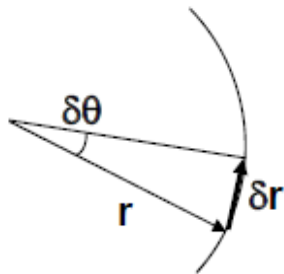
$$\underbrace{\frac{dA'_x}{dt} \hat{i}' + \frac{dA'_y}{dt} \hat{j}' + \frac{dA'_z}{dt} \hat{k}'}_{\left(\frac{d\vec{A}}{dt} \right)_{\text{inertial}}} = \underbrace{\frac{dA'_x}{dt} \hat{i}' + \frac{dA'_y}{dt} \hat{j}' + \frac{dA'_z}{dt} \hat{k}'}_{\left(\frac{d\vec{A}}{dt} \right)_{\text{rotating}}} + \underbrace{A'_x \frac{d\hat{i}'}{dt} + A'_y \frac{d\hat{j}'}{dt} + A'_z \frac{d\hat{k}'}{dt}}_{\text{effects of rotation}}$$

Total Differentiation in a Rotating System

$$\underbrace{\frac{dA_x}{dt}\hat{i} + \frac{dA_y}{dt}\hat{j} + \frac{dA_z}{dt}\hat{k}}_{\left(\frac{d\vec{A}}{dt}\right)_{inertial}} = \underbrace{\frac{dA'_x}{dt}\hat{i}' + \frac{dA'_y}{dt}\hat{j}' + \frac{dA'_z}{dt}\hat{k}'}_{\left(\frac{d\vec{A}}{dt}\right)_{rotating}} + \underbrace{A'_x\frac{d\hat{i}'}{dt} + A'_y\frac{d\hat{j}'}{dt} + A'_z\frac{d\hat{k}'}{dt}}_{\text{effects of rotation}}$$

To interpret $\frac{d\hat{i}'}{dt}, \frac{d\hat{j}'}{dt}, \frac{d\hat{k}'}{dt}$ think of each unit vector as a position vector.

linear velocity = angular velocity x position vector $\rightarrow \vec{V} = \vec{\Omega} \times \vec{r}$



Because $\vec{V} = \frac{d\vec{r}}{dt}$, $\frac{d\vec{r}}{dt} = \vec{\Omega} \times \vec{r}$

Thus $\frac{d\hat{i}'}{dt} = \vec{\Omega} \times \hat{i}', \quad \frac{d\hat{j}'}{dt} = \vec{\Omega} \times \hat{j}', \quad \frac{d\hat{k}'}{dt} = \vec{\Omega} \times \hat{k}'$

Total Differentiation in a Rotating System

$$\underbrace{\frac{dA_x}{dt}\hat{i} + \frac{dA_y}{dt}\hat{j} + \frac{dA_z}{dt}\hat{k}}_{\left(\frac{d\vec{A}}{dt}\right)_{inertial}} = \underbrace{\frac{dA'_x}{dt}\hat{i}' + \frac{dA'_y}{dt}\hat{j}' + \frac{dA'_z}{dt}\hat{k}'}_{\left(\frac{d\vec{A}}{dt}\right)_{rotating}} + \underbrace{A'_x(\vec{\Omega} \times \hat{i}') + A'_y(\vec{\Omega} \times \hat{j}') + A'_z(\vec{\Omega} \times \hat{k}')}_{\vec{\Omega} \times \vec{A} \text{ (effects of rotation)}}$$

$$\left(\frac{d\vec{A}}{dt}\right)_{inertial} = \left(\frac{d\vec{A}}{dt}\right)_{rotating} + \vec{\Omega} \times \vec{A}$$

This equation provides us with a formal way of expressing the balance of forces on a fluid parcel in a rotating coordinate system.

Momentum Equation in Rotating Coordinates

- In an inertial reference frame, Newton's second law of motion may be written symbolically as:

$$\left(\frac{d\vec{V}}{dt} \right)_{inertial} = \frac{\sum \vec{F}}{m}$$

- Substitute the position vector $\vec{r}$ for air parcel rotating on the earth

the transformation of the total derivative, to get:

$$\left(\frac{d\vec{r}}{dt} \right)_{inertial} = \left(\frac{d\vec{r}}{dt} \right)_{rotating} + \vec{\Omega} \times \vec{r}$$

Because velocity is the rate of change of the position vector with time, this can be rewritten:

$$\vec{V} = \frac{d\vec{r}}{dt} \Rightarrow \vec{V}_{inertial} = \vec{V} + \vec{\Omega} \times \vec{r}$$

Now substitute $\vec{V}_{inertial}$ into the transformation of the total derivative:

$$\left(\frac{d\vec{V}_{inertial}}{dt} \right)_{inertial} = \frac{d\vec{V}_{inertial}}{dt} + \vec{\Omega} \times \vec{V}_{inertial}$$

Momentum Equation in Rotating Coordinates

Substitute for $\vec{V}_{inertial}$ in $\left(\frac{d\vec{V}_{inertial}}{dt}\right)_{inertial} = \frac{d\vec{V}_{inertial}}{dt} + \vec{\Omega} \times \vec{V}_{inertial}$ to get

$$\left(\frac{d\vec{V}}{dt}\right)_{inertial} = \frac{d}{dt}(\vec{V} + \vec{\Omega} \times \vec{r}) + \vec{\Omega} \times (\vec{V} + \vec{\Omega} \times \vec{r})$$

[Here $\vec{\Omega} \times (\vec{\Omega} \times \vec{r}) = \vec{\Omega} \times (\vec{\Omega} \times \vec{R}) = -\Omega^2 \vec{R}$]

Using some vector identities and defining $\vec{R}$ as a vector perpendicular to the axis of rotation with magnitude equal to the distance to the axis of rotation:

$$\left(\frac{d\vec{V}}{dt}\right)_{inertial} = \frac{d\vec{V}}{dt} + 2\vec{\Omega} \times \vec{V} - \Omega^2 \vec{R}$$



Acceleration
following the
motion in an
inertial system



Rate of change
of relative
velocity following
the relative
motion in a
rotating
reference frame.



Coriolis
acceleration



Centrifugal
acceleration

Momentum Equation in Rotating Coordinates

Substituting into Newton's second law:

$$\left(\frac{d\vec{V}}{dt} \right)_{inertial} = \frac{\sum \vec{F}}{m}$$
$$\frac{d\vec{V}}{dt} + 2\vec{\Omega} \times \vec{V} - \Omega^2 \vec{R} = \frac{\sum \vec{F}}{m}$$

If the real forces acting on a fluid parcel are the pressure gradient force, gravitation and friction, then

The diagram shows the vector momentum equation in rotating coordinates: $\frac{d\vec{V}}{dt} = -2\vec{\Omega} \times \vec{V} - \frac{1}{\rho} \nabla p + \vec{g} + \vec{F}_r$. Red arrows point from descriptive text to each term in the equation:

- $\frac{d\vec{V}}{dt}$: Rate of change of relative velocity following the relative motion in a rotating reference frame.
- $-2\vec{\Omega} \times \vec{V}$: Coriolis acceleration
- $-\frac{1}{\rho} \nabla p$: Pressure gradient force (per unit mass)
- $\vec{g}$: Gravity term (gravitation + centrifugal)
- $\vec{F}_r$: Friction

Vector momentum equation in rotating coordinates

$$\vec{g} \equiv -g\vec{k} \equiv \vec{g}^* + \Omega^2 \vec{R}$$

The Laws Of Conservation Of Energy

The Thermodynamic Energy Equation

- The first law of thermodynamics is usually derived by considering a system in thermodynamic equilibrium, that is, a system that is **initially at rest** and **after exchanging heat with its surroundings and doing work on the surroundings is again at rest**. For such a system the first law states that *“the change in internal energy of the system is equal to the difference between the heat added to the system and the work done by the system”*.

❑ *A Lagrangian control volume consisting of a specified mass of fluid may be regarded as a thermodynamic system.* However, unless the fluid is at rest, it will not be in thermodynamic equilibrium.

❑ Nevertheless, the first law of thermodynamics still applies. To show that this is the case, we note that the total thermodynamic energy of the control volume is considered to consist of the sum of the internal energy (due to the kinetic energy of the individual molecules) and the kinetic energy due to the macroscopic motion of the fluid.

The Thermodynamic Energy Equation

- ❑ This law states that (1) heat is a form of energy that (2) its conversion into other forms of energy is such that total energy is conserved.

The diagram shows the equation $\Delta U = Q - W$ enclosed in a black rectangular box. Three arrows point from descriptive text labels below to the terms in the equation: one from 'change in internal energy (related to temperature)' to ΔU , one from 'Heat added to the system' to Q , and one from 'Work done by the system' to W .

$$\Delta U = Q - W$$

change in internal energy
(related to temperature)

Heat added to the system

Work done by the system

- ❑ *Each air parcel contains molecules that have internal energy*, which when thinking about the atmosphere, is just the *kinetic energy* of the molecules (associated with molecular rotations and, in some cases, vibrations) and *the potential energy* of the molecules (associated with the attractive and repulsive forces between the molecules).
- ❑ Internal energy does not consider their chemical bonds nor the nuclear energy of the nucleus because these do not change during collisions between air molecules.

□ **Doing work** on an air parcel involves either expanding it by increasing its volume or contracting it. *In the atmosphere, as in any system of molecules, energy is not created or destroyed, but instead, it is conserved.*

The thermodynamic energy equation is simply the statement of the first law of thermodynamics after taking into account the motion of the atmosphere. The first law of thermodynamics is written in the form

The Thermodynamic Energy Equation

- The first law of thermodynamics states that the change in internal energy e of a system is equal to the heat added to the system minus the work done by the system on its environment

$$de = dq - dw$$

- The thermodynamic energy equation

$$c_v \frac{DT}{Dt} + p \frac{D\alpha}{Dt} = J$$

- $c_v \frac{DT}{Dt}$ change in internal (thermal) energy of dry air
- J diabatic heating, the heating from external means
- $p \frac{D\alpha}{Dt}$ rate of work done by the fluid system on its environment

....END

Thanks !